Northern Hemisphere LGM Ice-Sheet Reconstruction with Yelmo Forced by CLIMBER-X

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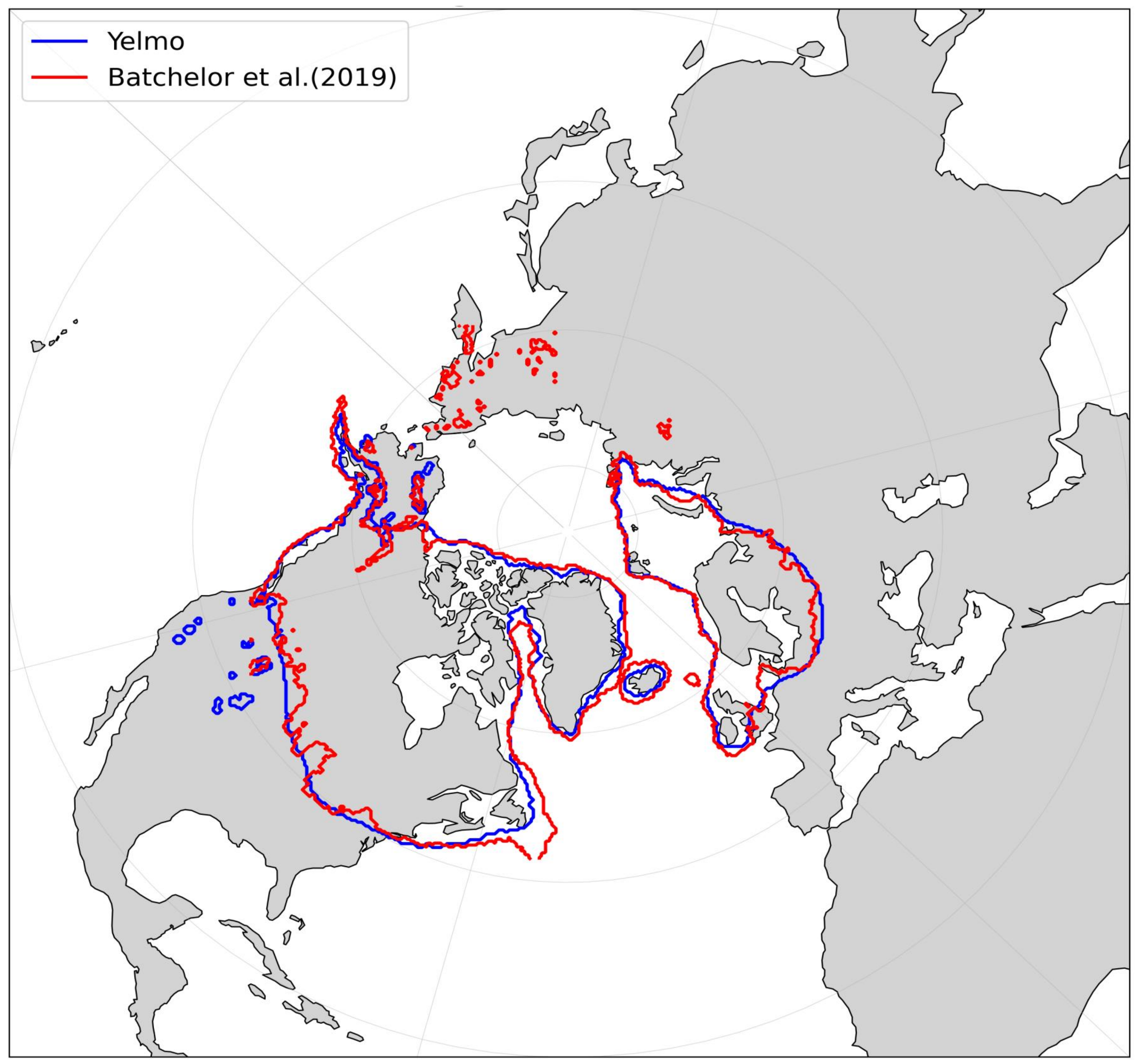
Goals

- Reconstruct NH ice sheets during the LGM using the ice-sheet model Yelmo interactively coupled with CLIMBER-X.
- Bridge the gap between observation-based ice-sheet reconstructions and physically consistent coupled climate–ice-sheet simulations.
- Evaluate the simulated LGM ice-sheet extent against the geomorphological reconstruction.
- Identify regions of agreement and mismatch to guide future improvements in coupled ice-sheet reconstruction workflows.

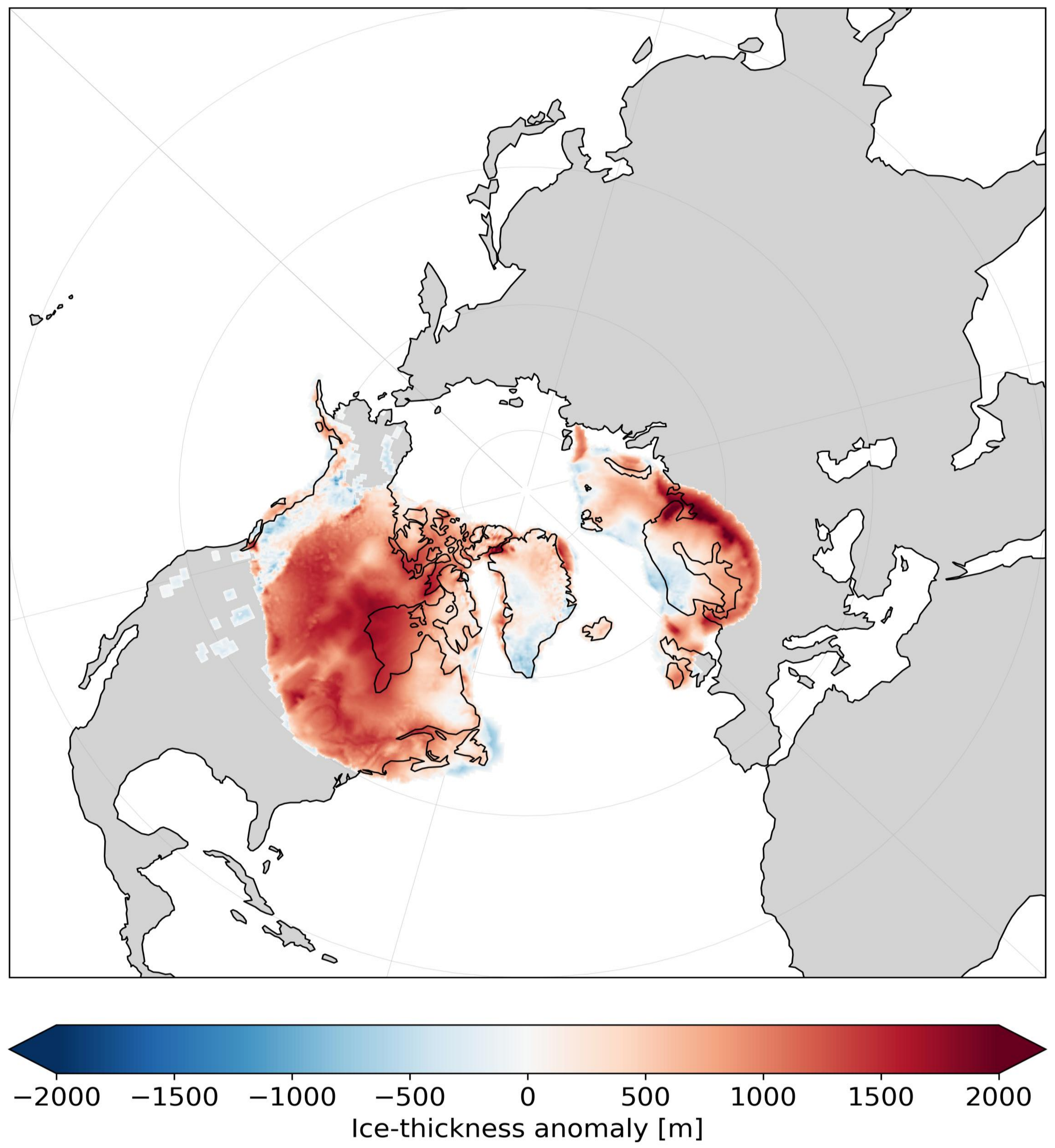
Simulation Setup

- Run an ensemble of simulations varying ice-sheet and solid-earth model parameters.
- Northern Hemisphere 32 km ice-sheet domain.
- 20 kyr simulation starting from LGM boundary conditions.
- Climate forcing every 10 years.
- Initialized from GLAC1D/Tarasov LGM geometry; Ice-sheet is then simulated freely by Yelmo.

LGM NH ice sheet margins: Yelmo vs Batchelor et al.(2019)



LGM ice-thickness: Yelmo minus GLAC1D



SMB Scheme

- The surface mass balance is parameterized using a prescribed snow-line elevation based on latitude, atmospheric CO_2 concentration, and orbital forcing.
- For each grid cell, SMB depends on the elevation difference between the surface elevation and the estimated snow line.
- Separate SMB equations are applied above and below the snow line, with an additional correction using the paleo-informed ice-margin mask from Batchelor et al. (2019).
- A relatively realistic, parameterized SMB scheme nudges ice growth toward the target LGM extent while allowing Yelmo to evolve dynamically.

Next Steps

- Tune SMB parameters to improve the match with reconstructed ice thickness.
- Test the sensitivity of the reconstructed ice sheets to different Earth viscosity structures.
- Extend the model–data comparison beyond ice extent by including sea-level and glacial isostatic adjustment constraints.

References: Batchelor et al. (2019), Nature communications, 10(1):3713; Ganopolski and Brovkin (2017), Climate of the Past, 13(12), 1695-1716; Robinson et al. (2020), Geoscientific Model Development, 13(6):2805–2823; Tarasov et al. (2012), Earth and Planetary ScienceLetters, 315:30–40; Willeit et al. (2022), Geoscientific Model Development, 15(14):5905–5948.